PROJECT DOCUMENTATION

Enterprise Cybersecurity & Network Security

Security Overview Dashboard in Power BI

Security incidents, network activity and vulnerability posture

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Type: Company training project

Tool: Microsoft Power BI (Desktop and Service)

1. Project Overview

This project analyses enterprise cybersecurity data in Power BI. It brings five raw CSV files (users, devices, security incidents, network logs and vulnerabilities) into one cleaned data model and presents the results in a four-page interactive report. The report helps a security team see how many incidents are happening, where and how they are detected, how network traffic is controlled, and how exposed the devices are to unpatched vulnerabilities.

Objectives

- Clean and standardise inconsistent raw data using Power Query.
- Build a star-style data model with a shared Date table and reusable DAX measures.
- Provide executive KPIs and drill-down analysis for incidents, network traffic and vulnerabilities.
- Publish the report to the Power BI Service for viewing and sharing.

Solution at a glance

Area	Details
Data sources	5 CSV files: Users, Devices, Security_Incidents, Network_Logs, Vulnerabilities
Data model	7 tables: 5 data tables, a Date table and a Measure table; 7 one-to-many relationships
Report	4 pages at 1920 x 1080: Security Overview, Network Analytics, Threat Analysis, Risk & Vulnerabilities
Interactivity	Page navigator, slicers, drill-down hierarchies, cross-filtering and Reset Filters bookmarks
Time span	January 2025 to December 2026 (730-row Date table)
Delivery	Published to the Power BI Service (My workspace) as "Project1"

2. Data Sources

All data was loaded from CSV files. Row counts below are the full table sizes; the Power Query column profiles in Section 3 are based on the first 1,000 rows only.

Table	Rows	Columns	Description and fields
Users	300	6	Employee master. UserID, EmployeeCode, UserName, Department, Role, Location
Devices	150	8	Asset inventory. DeviceID, AssetTag, DeviceType, OperatingSystem, Department, Location, Criticality, Status
Security_Incidents	1,200	10	Incident records. IncidentID, IncidentDate, DeviceID, UserID, AttackType, Severity, Status, DetectionSource, DetectionTime, ResolutionTime
Network_Logs	2,500	10 (+1 added)	Traffic logs. LogID, LogDate, DeviceID, SourceIP, DestinationIP, DestinationPort, Protocol, Action, BytesTransferred, PacketCount; BytesTransferred_Clean added in Power Query
Vulnerabilities	500	9	Vulnerability findings. VulnerabilityID, DeviceID, CVE, VulnerabilityType, RiskScore, RiskLevel, DiscoveredDate, PatchDate, PatchStatus

3. Data Cleaning and Preparation (Power Query)

The raw files contained typical quality problems: leading and trailing spaces, mixed upper and lower case, misspelled or abbreviated labels, empty values and negative byte counts. These split one real category into several (for example "high", "High" and "HIGH"), which would give wrong counts in slicers and charts. Every table was cleaned in the Power Query Editor, and each step stays in the Applied Steps list so the process can be repeated on refresh.

Standard techniques used

- **Use First Row as Headers** on every CSV so column names are correct.
- **Transform > Trim** to remove hidden leading and trailing spaces.
- **Transform > Capitalize Each Word / UPPERCASE** to make text values consistent.
- **Replace Values** to fix specific wrong labels and to turn empty strings into "Unknown".
- **Custom column** with a conditional rule to neutralise invalid numbers.

3.1 Users

Column	Problem found	Fix applied	Result
(all)	Header sat in the first data row	Use First Row as Headers	Correct column names
Department	Case and spacing variants such as "it", "Finance" and "R&D" duplicates (11 distinct values)	Trim; replace "it" with "IT"	8 clean departments
Role	Lowercase variants such as "developer" and "manager" (9 distinct values)	Trim; Capitalize Each Word	7 clean roles
Location	1 empty value	Replace empty string with "Unknown"	100% valid; 7 distinct (6 cities + Unknown)

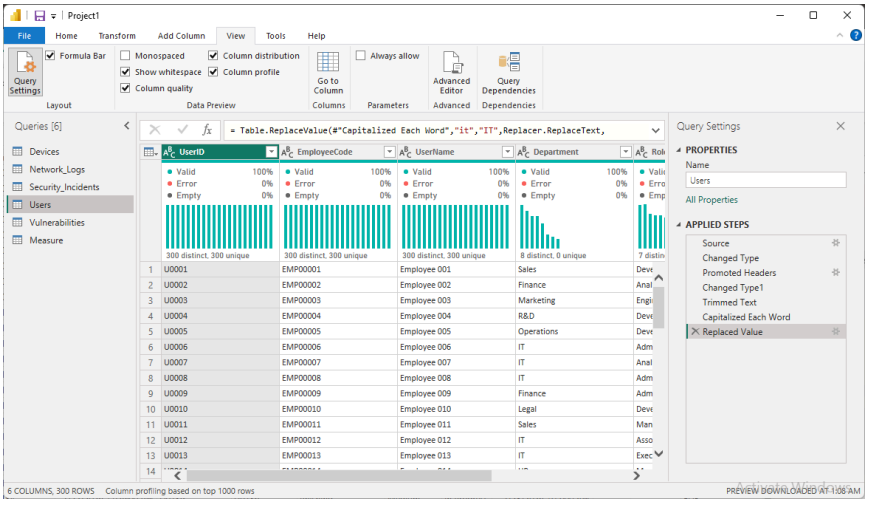


Figure 1: Users query: Applied Steps in Power Query

3.2 Devices

Column	Problem found	Fix applied	Result
(all)	Header sat in the first data row	Use First Row as Headers	Correct column names
OperatingSystem	Inconsistent labels such as "Win11" and "ubuntu" (12 distinct values)	Replace Win11 with Windows 11 and ubuntu with Ubuntu Linux; Trim	9 clean operating systems
Criticality	Case variants such as "high" and "Critical" duplicates (6 distinct values)	Trim; Capitalize Each Word; replace high with High	4 values: Low, Medium, High, Critical
Department	Needed to match the Users table so the two can be compared	Trim; Capitalize Each Word	8 departments, consistent with Users

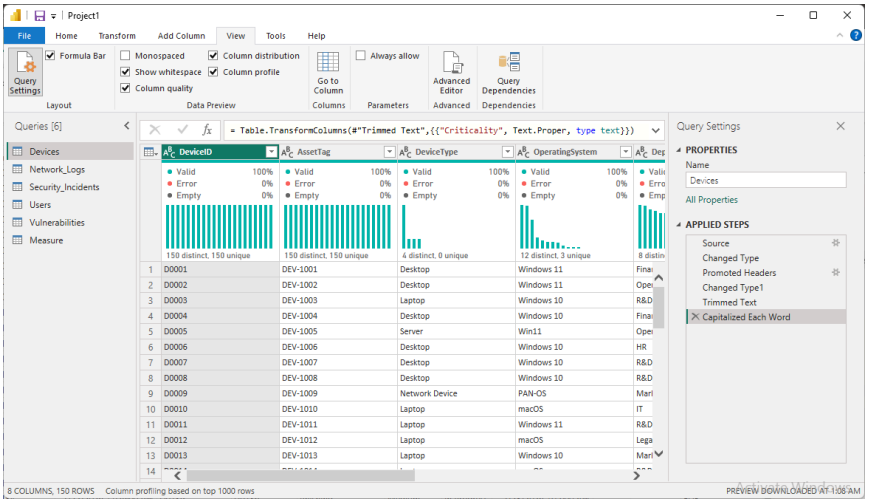


Figure 2: Devices query: Applied Steps in Power Query

3.3 Security_Incidents

Column	Problem found	Fix applied	Result
Severity	Mixed case such as "HIGH" and "high" (7 distinct values)	Trim; Capitalize Each Word	4 values: Low, Medium, High, Critical
Status	Variants such as "open" and "In-progress" (6 distinct values)	Trim; Capitalize Each Word; replace In-progress with In Progress	4 values: Open, In Progress, Resolved, Closed
DetectionSource	1 empty value and a duplicate "Email Gateway" caused by a trailing space (8 distinct values)	Trim; replace empty string with "Unknown"	7 distinct (6 sources + Unknown), 100% valid

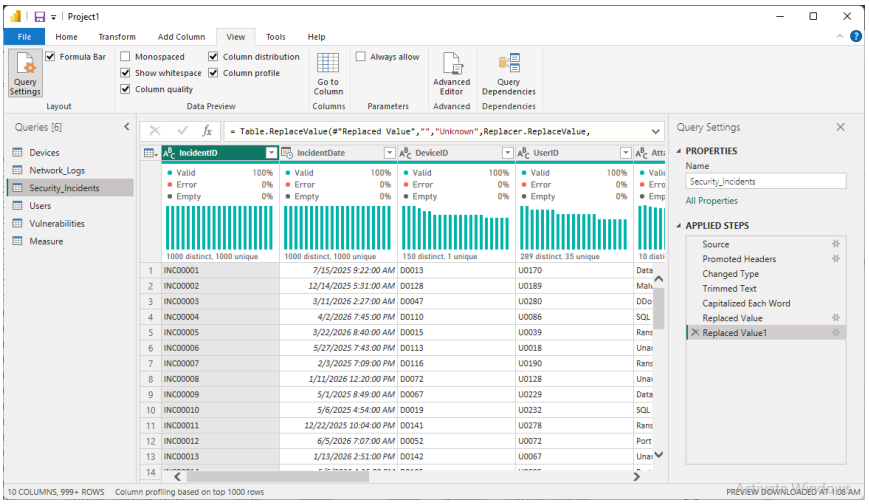


Figure 3: Security_Incidents query: Applied Steps in Power Query

3.4 Network_Logs

Column	Problem found	Fix applied	Result
Protocol	Lowercase "tcp" counted separately from "TCP" (9 distinct values)	Trim; UPPERCASE	8 protocols: HTTP, DNS, FTP, HTTPS, UDP, TCP, SMTP, SSH
Action	Variants such as "blocked" (5 distinct values)	Trim; Capitalize Each Word	3 values: Allowed, Blocked, Suspicious
BytesTransferred	Negative values, which are impossible for bytes	Custom column BytesTransferred_Clean: if [BytesTransferred] < 0 then null else [BytesTransferred]	Negatives become null so they do not distort Total Bytes

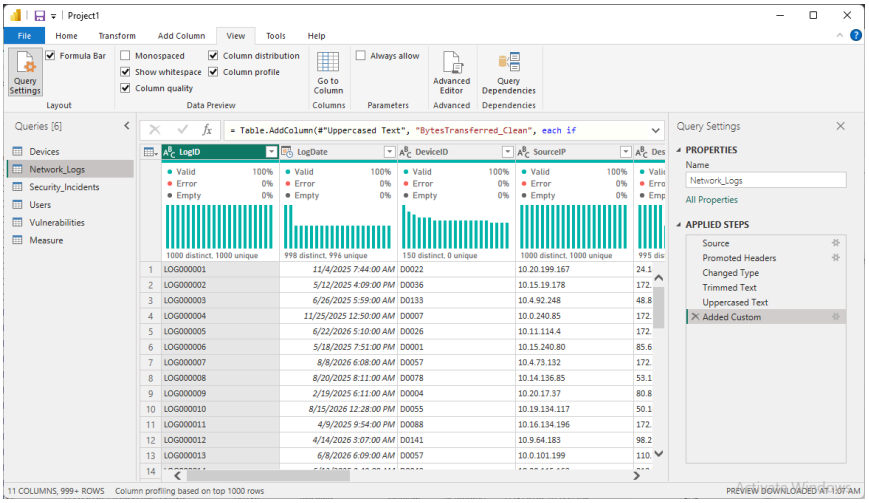


Figure 4: Network_Logs query: Applied Steps in Power Query

3.5 Vulnerabilities

Column	Problem found	Fix applied	Result
RiskLevel	Case variants such as "high" (6 distinct values)	Trim; Capitalize Each Word	4 values: Low, Medium, High, Critical
PatchStatus	Variants such as "pending" and "not patched" (6 distinct values)	Trim; Capitalize Each Word	4 values: Patched, Pending, Not Patched, Risk Accepted
CVE	About 40% of rows have no CVE ID	Intentionally kept as null	Not every weakness has a CVE, so nothing was invented

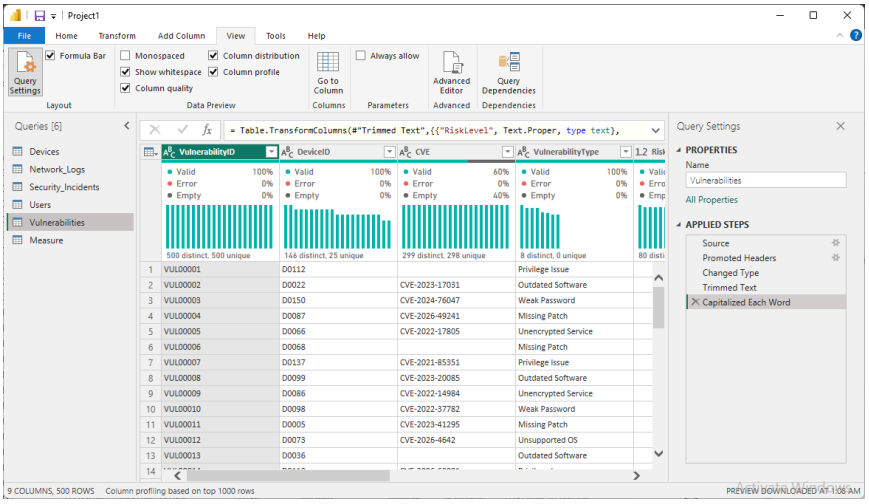


Figure 5: Vulnerabilities query: Applied Steps in Power Query

3.6 Date table and Measure table

The Date table was created inside the model and covers every day from 1 January 2025 to 31 December 2026 (730 rows). The Measure table holds only DAX measures and no raw data, so it needed no cleaning.

4. Data Model

The model follows a star-schema pattern. Devices, Users and Date act as dimension tables that filter the three fact tables: Security_Incidents, Network_Logs and Vulnerabilities. Key columns used only for joining (DeviceID, UserID) are hidden from the report view to keep the field list clean.

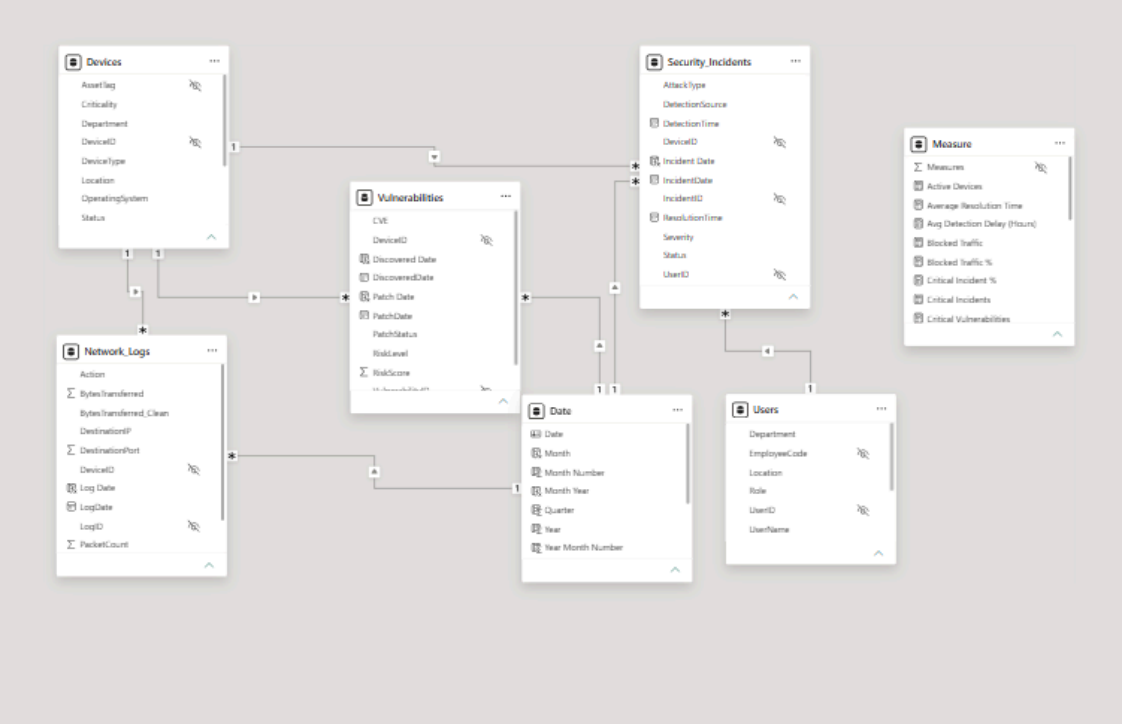


Figure 6: Model view showing tables and relationships

4.1 Table roles

Table	Role	Notes
Users	Dimension	Who is affected by incidents; has the Organization hierarchy
Devices	Dimension	Device inventory; has the Device hierarchy
Date	Dimension	Calendar table shared by all fact tables
Security_Incidents	Fact	One row per incident
Network_Logs	Fact	One row per network log entry
Vulnerabilities	Fact	One row per vulnerability finding
Measure	Measure table	Stores all DAX measures in display folders

4.2 Relationships

All relationships are one-to-many, with the filter flowing from the dimension (one side) to the fact table (many side).

From (one)	To (many)	Joined on
Devices	Security_Incidents	DeviceID
Devices	Network_Logs	DeviceID
Devices	Vulnerabilities	DeviceID
Users	Security_Incidents	UserID
Date	Security_Incidents	Incident date
Date	Network_Logs	Log date

From (one)	To (many)	Joined on
Date	Vulnerabilities	Discovered date

4.3 Hierarchies

Hierarchy	Table	Levels
Organization	Users	Location > Department > Role > UserName
Device	Devices	Location > Department > DeviceType > DeviceID
Date Hierarchy	Date	Year > Quarter > Month > Day

The Organization and Device hierarchies power the drill-down charts described in Section 7.

4.4 Date table

Columns in the Date table: Date, Year, Month Number, Month, Month Year, Year Month Number and Quarter, plus a Date Hierarchy. Month Year is used on the Network Analytics trend chart and Year is used in the Year slicers.

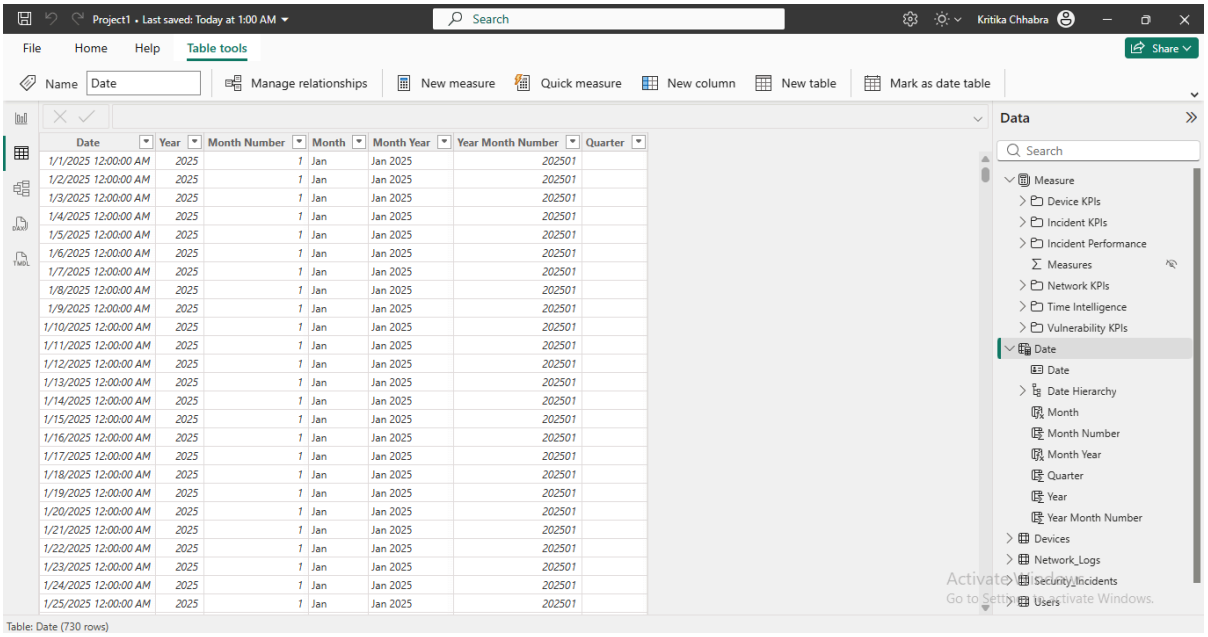


Figure 7: Date table in Table view (730 rows)

5. DAX Measures

All measures live in a dedicated Measure table and are grouped into display folders: Device KPIs, Incident KPIs, Incident Performance, Network KPIs, Time Intelligence and Vulnerability KPIs. Reports use measures, never raw column sums, so every page shares the same definitions.

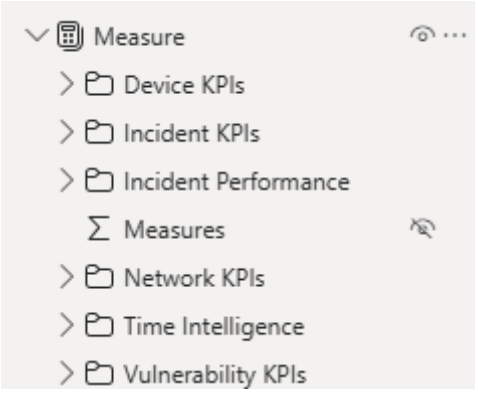


Figure 8: Measure table with its display folders

Measure	What it shows	Used on
Total Incidents	Number of security incidents	Overview KPI; Threat Analysis matrix, location and detection-source charts
Critical Incidents	Incidents with Critical severity (74 overall)	Overview KPI
Open Incidents	Incidents that are still open (167 overall)	Overview KPI
Critical Incident %	Critical incidents as a share of all incidents	Available for analysis
Average Resolution Time	Average time taken to resolve an incident (50.3 overall)	Overview KPI; Threat Analysis by attack type
Avg Detection Delay (Hours)	Average delay before an incident is detected, in hours	Available for analysis
Total Vulnerabilities	Number of vulnerability findings (500 overall)	Overview KPI; all Risk & Vulnerabilities charts
Critical Vulnerabilities	Findings with Critical risk level (52 overall)	Overview KPI
Total Network Logs	Number of network log entries (2,500 overall)	Network KPI card, trend, protocol and action charts
Blocked Traffic	Log entries with action Blocked (270)	Network KPI card
Blocked Traffic %	Blocked traffic as a share of all traffic (10.8%)	Network KPI card
Suspicious Traffic	Log entries with action Suspicious (135)	Network KPI card
Total Bytes	Sum of cleaned bytes transferred (about 6M)	Network KPI card
Active Devices	Number of devices in service	Available for analysis

6. Report Design

The report uses a custom purple theme on top of the Fluent 2 base theme, a Segoe UI heading style, and a light cream canvas. Every page is 1920 x 1080 and has the same header area (title, subtitle) and a page navigator across the top so users can move between pages with one click.

6.1 Security Overview

A one-glance executive summary of incidents and vulnerabilities.

- Six KPI cards: Total Incidents, Critical Incidents, Open Incidents, Total Vulnerabilities, Critical Vulnerabilities and Average Resolution Time.



Figure 9: Security Overview page

6.2 Network Analytics

Monitors traffic volume, how much is controlled, and which protocols are used.

- Five KPI cards: Blocked Traffic, Blocked Traffic %, Suspicious Traffic, Total Bytes and Total Network Logs.
- Line chart of Total Network Logs by Month Year, donut chart by Action, and column chart by Protocol.

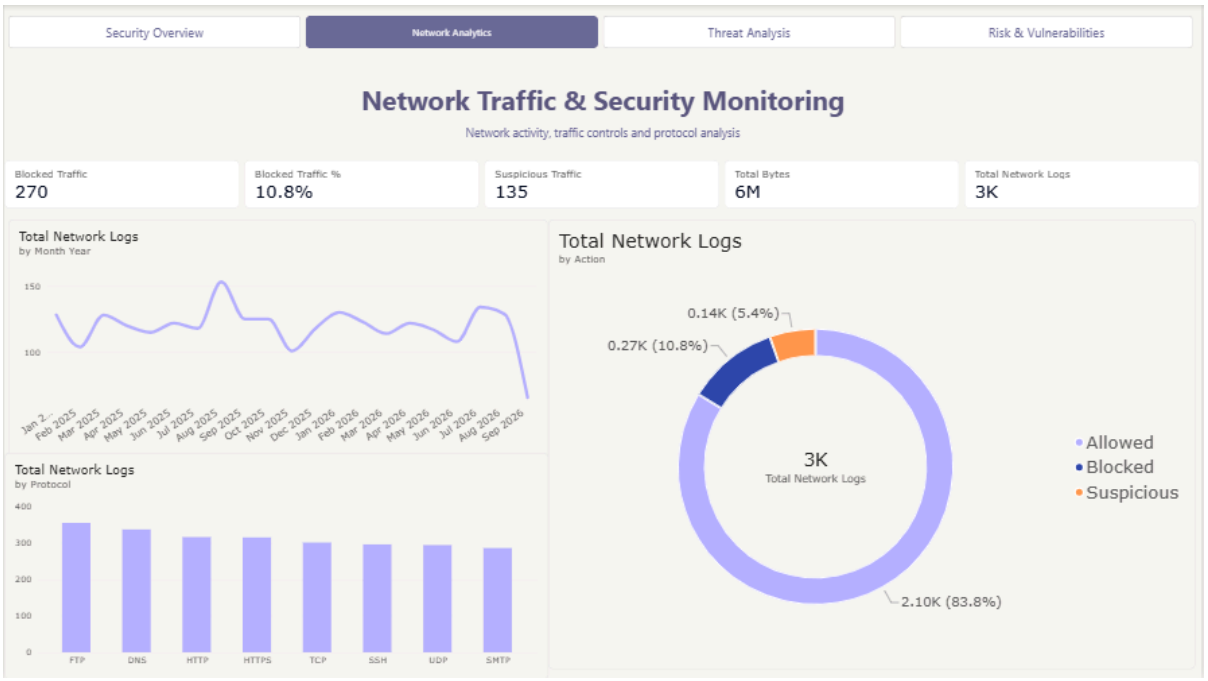


Figure 10: Network Analytics page

6.3 Threat Analysis

Detailed analysis of attack patterns, severity and resolution.

- Matrix of AttackType by Severity showing Total Incidents.
- Column chart of Total Incidents by the Organization hierarchy (starting at Location), bar chart of Total Incidents by DetectionSource, and column chart of Average Resolution Time by AttackType.
- Slicers: Year, AttackType, Severity and Status, plus a Reset Filters button.

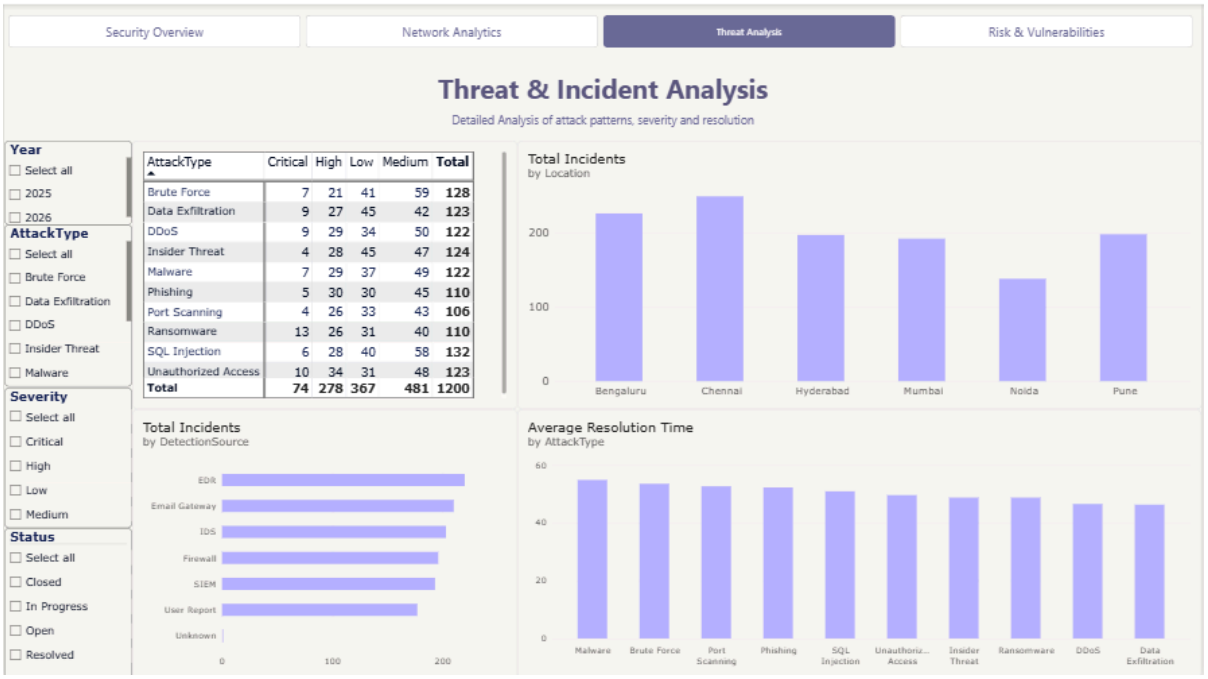


Figure 11: Threat Analysis page

6.4 Risk & Vulnerabilities

Shows device exposure, vulnerability severity and patch posture.

- Donut chart of Total Vulnerabilities by RiskLevel and pie chart by PatchStatus.
- Bar chart by Department and column chart by the Device hierarchy (starting at Location).
- Slicers: Year, Department, PatchStatus and RiskLevel, plus a Reset Filters button.



Figure 12: Risk & Vulnerabilities page

7. Interactivity and Navigation

7.1 Page navigation and Reset Filters

A page navigator on every page switches between the four pages. The Threat Analysis and Risk & Vulnerabilities pages each have a Reset Filters button that uses a bookmark to clear all slicers and selections in one click.

7.2 Slicers and cross-filtering

Slicers filter all visuals on their page. Clicking a bar, slice or matrix cell also cross-filters the other visuals. The examples below show the Threat Analysis page filtered to Year 2025, and then to Year 2025, Data Exfiltration, High severity and Closed status, which leaves 2 incidents.

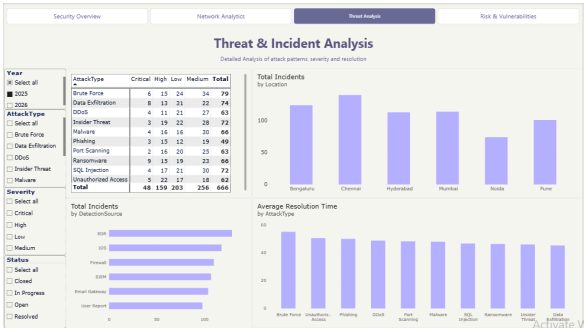


Figure 13: Threat Analysis with the Year 2025 slicer selected

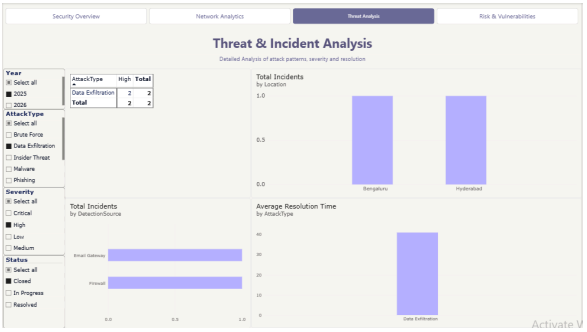


Figure 14: Threat Analysis with Year, AttackType, Severity and Status selected



Figure 15: Risk & Vulnerabilities filtered to Year 2025

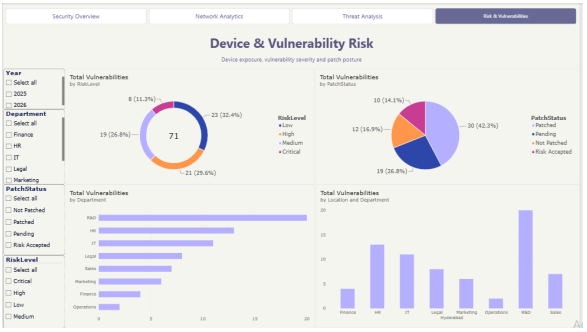


Figure 16: Cross-filtering after selecting a bar in the location chart

7.3 Drill-down

The location charts are built on hierarchies, so a user can drill down one level at a time using the drill arrows on the visual.

Risk & Vulnerabilities: Location > Department > DeviceType > DeviceID

Example path: Hyderabad > R&D > Laptop > device IDs D0003 and D0014. The other visuals update at each level.

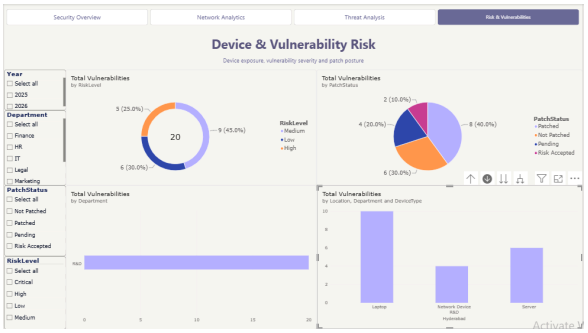


Figure 17: Drilled to DeviceType (Hyderabad, R&D)

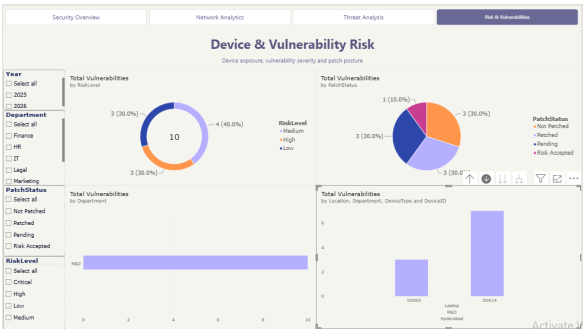


Figure 18: Drilled to DeviceID (Laptop)

Threat Analysis: Location > Department > Role > UserName

Example path: Chennai > IT > Engineer > individual employees (Employee 043, 189, 267 and 280).

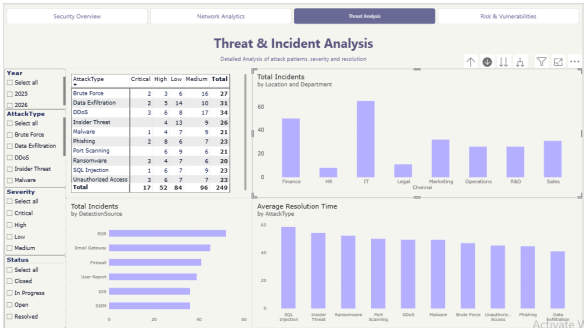


Figure 19: Drilled to Department (Chennai)

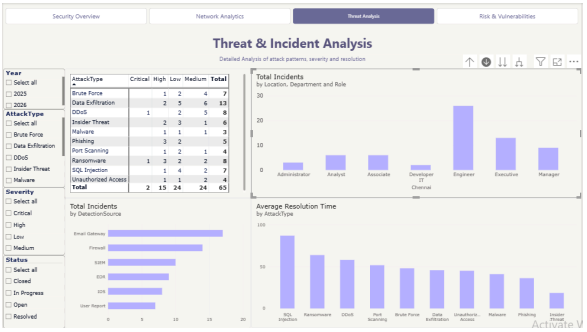


Figure 20: Drilled to Role (Chennai, IT)

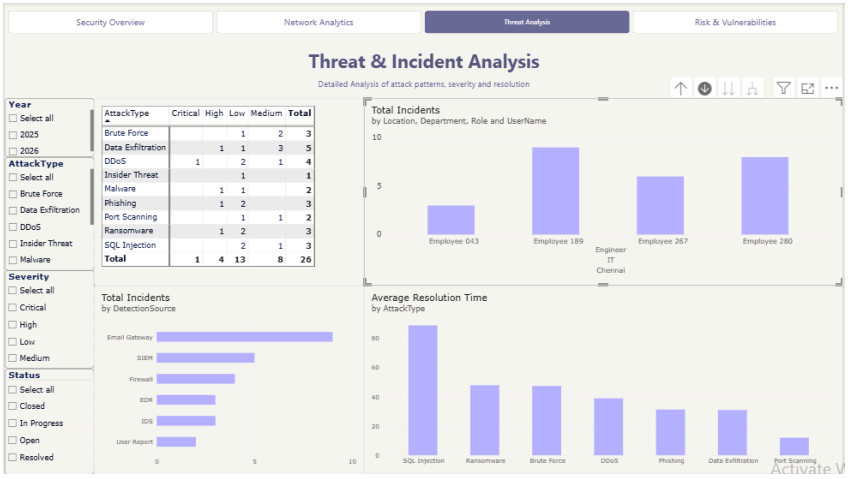


Figure 21: Drilled to UserName (Chennai, IT, Engineer)

8. Key Insights

The observations below come from the unfiltered report (all years).

Incidents

- **1,200 incidents** were recorded; 74 are Critical (about 6%) and 167 are still open.
- By severity the split is Medium 481, Low 367, High 278 and Critical 74.
- SQL Injection is the most frequent attack type (132), followed by Brute Force (128). Ransomware has the most Critical incidents (13).
- Chennai has the most incidents and Noida the fewest. EDR is the top detection source.
- Malware takes the longest to resolve on average; Data Exfiltration takes the shortest. The overall average resolution time is 50.3.

Network traffic

- Of 2,500 logs, 83.8% were allowed, 10.8% (270) blocked and 5.4% (135) flagged as suspicious.
- Traffic is spread fairly evenly across the eight protocols, with FTP highest and SMTP lowest.
- Monthly volume peaks in September 2025. The sharp drop at the end of the line most likely reflects an incomplete final month.

Vulnerabilities

- 500 vulnerabilities, of which 52 (10.4%) are Critical and 124 (24.8%) High.
- Only 47.8% are patched. 23.6% are pending, 17.8% are not patched and 10.8% have been formally risk-accepted.
- Marketing has the most vulnerabilities by department and Operations the fewest. By location, Pune is highest and Mumbai lowest.

Recommendations

- Prioritise patching for the 89 Not Patched and 118 Pending items, starting with Critical and High risk.
- Review detection and response for SQL Injection and Brute Force, the two most frequent attack types.
- Investigate why Malware incidents take longest to resolve.

9. Publishing

The report was published from Power BI Desktop to the Power BI Service (My workspace) under the name Project1. The service view keeps the page list on the left, and users can subscribe, set alerts, share, or open the semantic model from the toolbar.

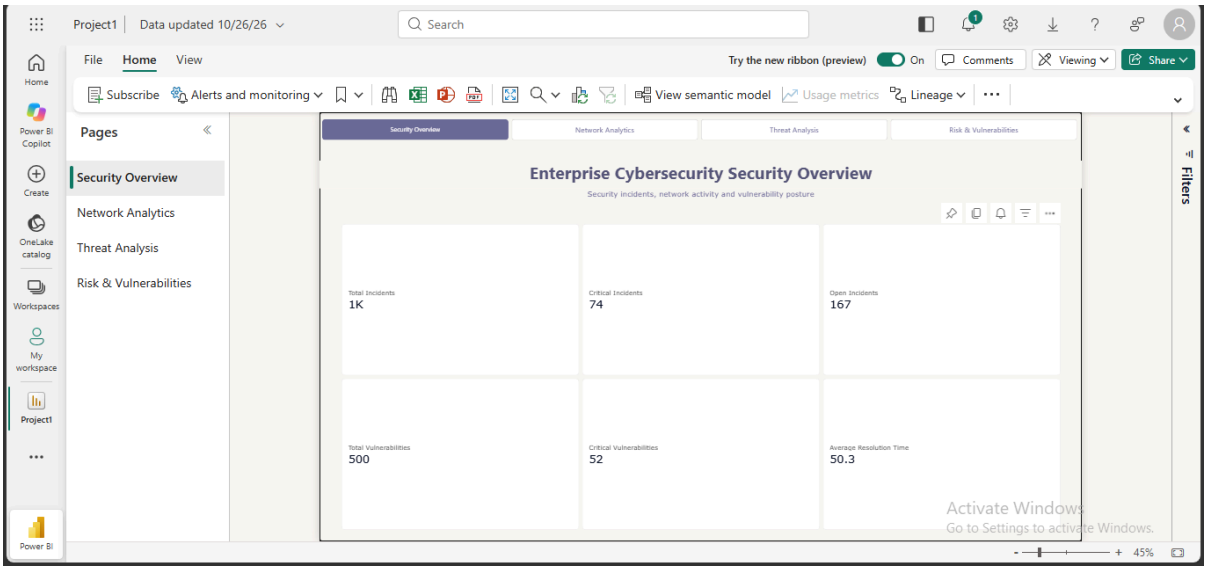


Figure 22: Published report in the Power BI Service

10. Summary

Raw, inconsistent CSV data was cleaned in Power Query, joined in a star-style model with a shared Date table, measured with reusable DAX, and presented as a four-page interactive dashboard that answers three questions: what is happening (incidents), how is traffic behaving (network), and how exposed are we (vulnerabilities).